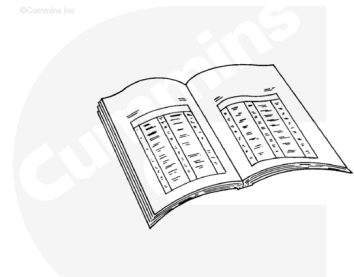


Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



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⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ CAUTION ⚠

Place a wooden block, the width of the oil pan, between the floor jack and oil pan to prevent damage to the engine.

- Remove the transmission, clutch, and all related components, if equipped. Refer to the OEM service manual.
- Remove the flywheel. Refer to Procedure 016-005 in Section 16. (</qs3/pubsys2/xml/en/procedures/35/35-016-005-tr.html>)
- Use a floor jack or a suitable lifting fixture to support the rear of the engine.
- Disconnect the batteries. Refer to Procedure 013-009 in Section 13. (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>)
- Remove the starting motor. Refer to Procedure 013-020 in Section 13. (</qs3/pubsys2/xml/en/procedures/35/35-013-020-tr.html>)
- Remove both rear engine mounts. Refer to Procedure 016-010 in Section 16.

(/qs3/pubsys2/xml/en/procedures/00/00-016-010.html)

- Remove the four oil pan rear mounting capscrews. Refer to Procedure 007-025 in Section 7.

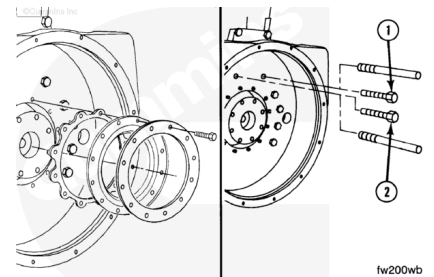
(/qs3/pubsys2/xml/en/procedures/101/101-007-025-tr.html)

- Remove the rear crankshaft oil seal. Refer to Procedure 001-024 in Section 1.

(/qs3/pubsys2/xml/en/procedures/101/101-001-024-tr.html)

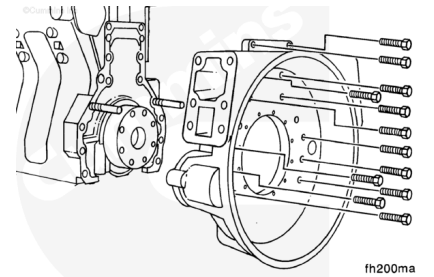
Remove

Remove capscrews 1 and 2 and install two guide pins, Part Number 3376697.



⚠ WARNING ⚠

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Remove the remaining capscrews.

Use a rubber hammer to loosen the flywheel housing.

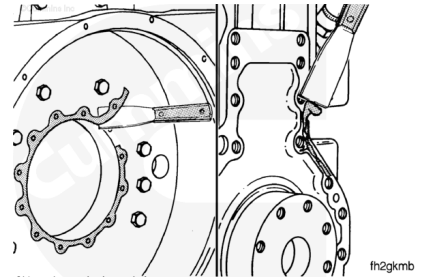
Do **not** damage the rear of the oil pan gasket when removing the flywheel housing.

Remove the flywheel housing.

Clean and Inspect for Reuse

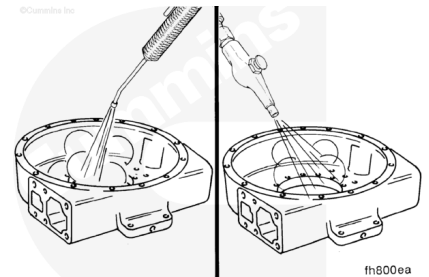
Remove the gasket material from the flywheel housing oil seal gasket surface.

Clean the sealant from the cylinder block surface.



⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

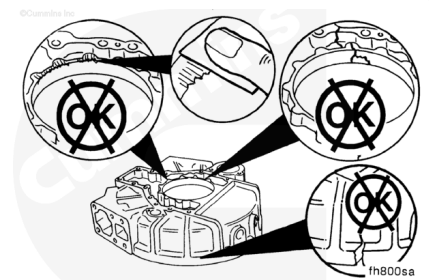
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use steam or solvent to clean the flywheel housing. Dry with compressed air.

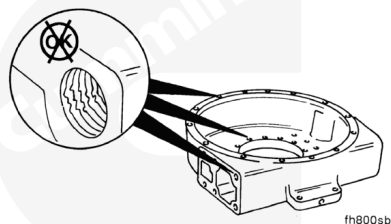
Inspect all surfaces for nicks, burrs, or cracks.

Use fine crocus cloth to remove small nicks and burrs.



Inspect all threaded capscrew holes for damage.

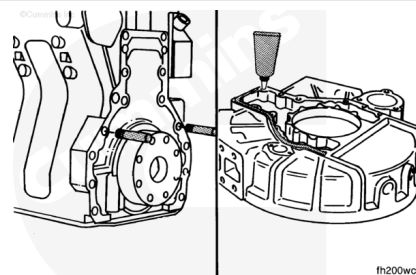
Repair or replace the housing if the capscrew holes are damaged.



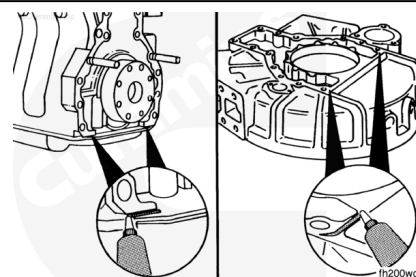
Install

Install two guide pins, Part Number 3376697, in the cylinder block to help support and align the housing during installation.

Apply a 2 mm [1/16 in] bead of silicone sealant, Part Number 3823494, on the mounting surface of the housing and around each capscrew hole.

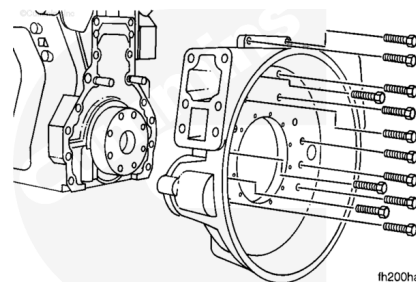


Apply silicone sealant to each rear corner of the oil pan gasket and flywheel housing mating surface.



⚠ WARNING ⚠

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Install the flywheel housing over the guide pins.

Do **not** damage the oil pan gasket when installing the flywheel housing.

The flywheel housing bore **must** be aligned with the crankshaft. Do **not** tighten the capscrews to the final torque value until the flywheel housing is aligned.

Install the capscrews. Remove the guide pins, and install the remaining two capscrews.

Torque Value: 7 n•m [60 in-lb]

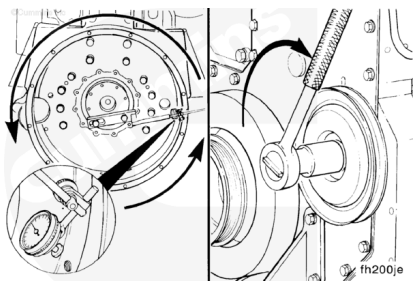
Measure

Bore Alignment

Use dial indicator gauge, Part Number 3376050, and dial gauge attachment, Part Number ST-1325, to measure the bore alignment.

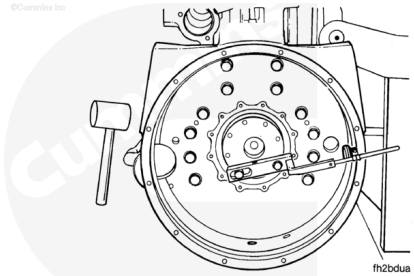
Adjust the dial indicator gauge to zero (0).

Use the accessory driveshaft to rotate the crankshaft one complete revolution (360 degrees).



Flywheel Housing Bore Alignment Maximum Total Indicator Reading

mm	SAE Number	in
0.30	00	0.012
0.25	0	0.010
0.25	1/2	0.010
0.20	1	0.008
0.20	2	0.008
0.20	3	0.008



If the maximum bore alignment does **not** meet the specifications, use a rubber hammer to move the housing in the necessary direction, and repeat the measurement procedure.

If the bore alignment will **not** meet the specifications or the bore is **not** round, the housing **must** be replaced.

Face Alignment

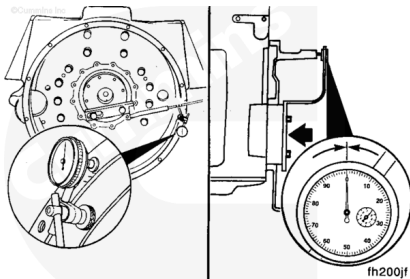
⚠ CAUTION ⚠

The tip of the gauge must not enter the capscrew holes or the gauge will be damaged.

Install the contact tip of the dial indicator gauge against the flywheel housing face.

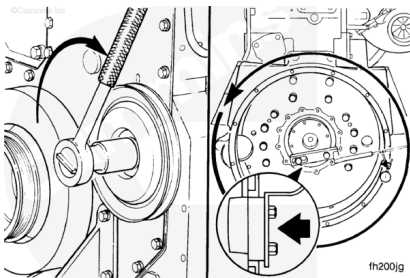
Push the crankshaft toward the front of the engine.

Adjust the dial indicator gauge to zero (0).



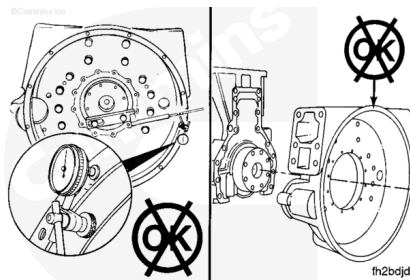
Monitor the dial indicator gauge while turning the crankshaft.

Use the accessory driveshaft to rotate the crankshaft one complete revolution (360 degrees) while keeping the crankshaft end thrust pushed forward.



Flywheel Housing Face Alignment Maximum Total Indicator Reading

mm	SAE Number	in
0.30	00	0.012
0.25	0	0.010
0.25	1/2	0.010
0.20	1	0.008
0.20	2	0.008
0.20	3	0.008



If the maximum face alignment does **not** meet the specifications, the flywheel housing **must** be replaced.

Tighten all capscrews in the sequence shown.

Torque Value:

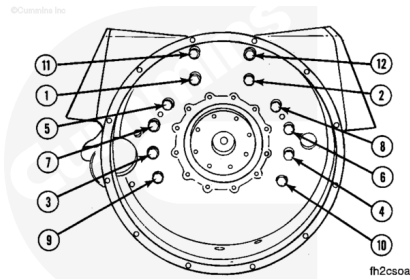
1. 68 n•m [50 ft-lb]

Torque Value:

1. 129 n•m [95 ft-lb]

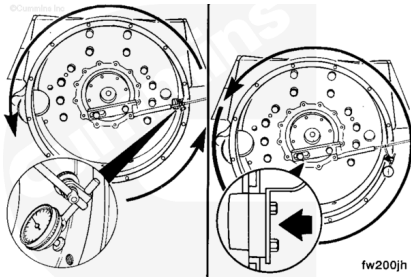
Torque Value:

1. 197 n•m [145 ft-lb]



Measure the bore and face alignment again.

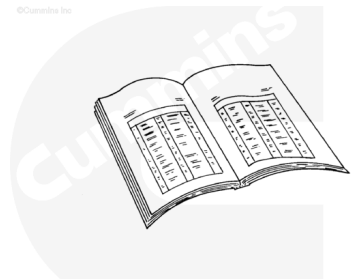
If bore and face alignment does **not** meet the specifications, loosen the housing capscrews. Tighten the capscrews, and measure the bore and face alignment again.



Finishing Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



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⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ CAUTION ⚠

Place a wooden block, the width of the oil pan, between the floor jack and oil pan to prevent damage to the engine.

- Install the rear crankshaft oil seal. Refer to Procedure 001-024 in Section 1.
(/qs3/pubsys2/xml/en/procedures/101/101-001-024-tr.html)
- Install the four oil pan rear mounting capscrews. Refer to Procedure 007-025 in Section 7.
(/qs3/pubsys2/xml/en/procedures/101/101-007-025-tr.html)
- Install both rear engine mounts. Refer to Procedure 016-010 in Section 16.
(/qs3/pubsys2/xml/en/procedures/00/00-016-010.html)

- Install the starting motor. Refer to Procedure 013-020 in Section 13. (</qs3/pubsys2/xml/en/procedures/35/35-013-020-tr.html>)
- Connect the batteries. Refer to Procedure 013-009 in Section 13. (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>)
- Remove the floor jack or lifting fixture used to support the rear of the engine.
- Install the flywheel. Refer to Procedure 016-005 in Section 16. (</qs3/pubsys2/xml/en/procedures/35/35-016-005-tr.html>)
- Install the transmission, clutch, and all related components, if equipped. Refer to the OEM service manual.
- Operate the engine and check for proper operation.